Question

You can write your answers in the provided space or write your answer on a piece of paper and scan and upload the handwritten answers using the QR Code available in the Scan and Upload Section of this exam.

Kindly ensure all answer sheets to be uploaded against the corresponding questions only. All sheets should not be uploaded against one question.

Recall, Laplace distribution is given by

$$f_X(x|\mu,\sigma) = \frac{1}{\sqrt{2}\sigma} e^{-\frac{\sqrt{2}|x-\mu|}{\sigma}}, x \in \mathbb{R}$$

If data instances {2, 1, 0, -1, -3} are obtained via sampling the above distribution, find the maximum likelihood estimate of (μ, σ).

[2.5+2.5 = 5 Marks]

Q. $f_X(x|\mu,\sigma) = \frac{1}{\sqrt{2}\,\sigma} e^{-\sqrt{2}\frac{|x-\mu|}{\sigma}}$

Data = $\{2, 1, 0, -1, -3\}$

find MLE (μ, σ)

Soln. MLE of μ.

For Laplacian distribution, MLE of μ is sample Median.

$\xrightarrow{\text{ordering data}}$ $\{-3, -1, 0, 1, 2\}$

$\xrightarrow{\text{Median}}$ $\boxed{\hat{\mu} = 0}$

MLE of σ.

$$\hat{\sigma} = \frac{\sqrt{2}}{n} \sum_{i=1}^{n} |x_i - \hat{\mu}|$$

Since, $\hat{\mu} = 0$ $\quad \sum x_i = |2| + |1| + |0| + |-3| + |-1|$

$\sum x_i = 7$

$n = 5$ So, $\boxed{\hat{\sigma} = \frac{\sqrt{2} \times 7}{5} = \frac{7\sqrt{2}}{5}}$